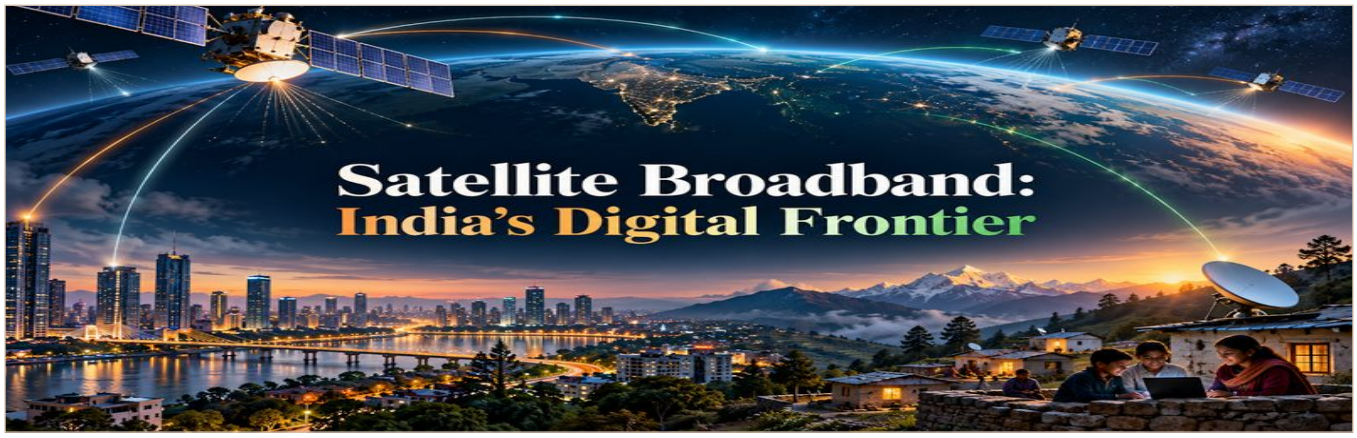


Akella Raghavendra Sir
Founder & Chief Mentor

[GS PAPER III: SCIENCE AND TECHNOLOGY; INTERNAL SECURITY. • GS PAPER II: GOVERNANCE.]
Publication Date: 21/10/2026

Satellite Broadband: India's Digital Frontier



Explore how satellite broadband can connect remote communities and bridge India's digital divide. Understand the affordability, competition and security challenges shaping India's digital frontier.

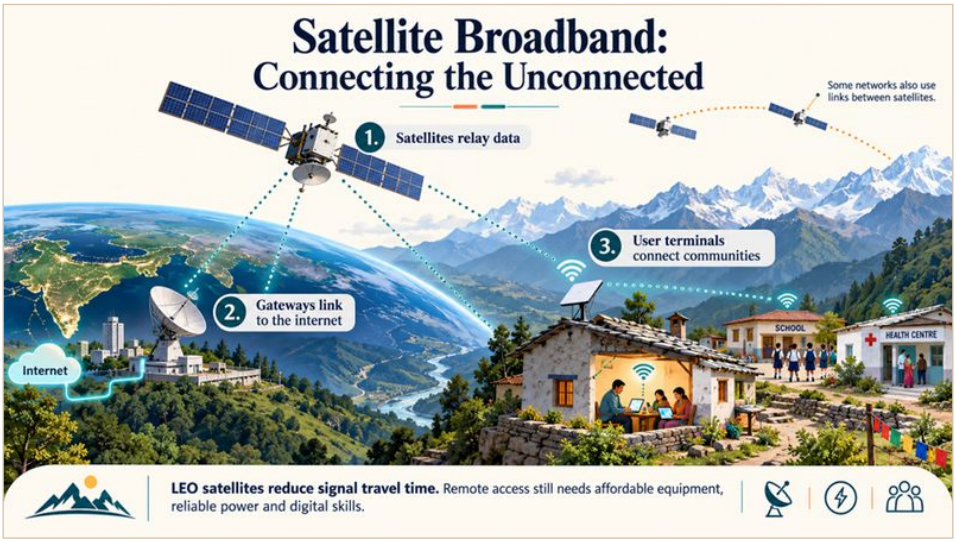
Satellite Broadband: India's Digital Frontier

Syllabus areas:

GS Paper III: Science and Technology; Internal Security.

GS Paper II: Governance.

Satellite broadband connects space technology with digital inclusion, commercial investment and national security. The expansion of services such as Starlink highlights an important policy challenge for India: extending affordable internet access while ensuring fair competition, reliable services and regulatory oversight.



What Is Satellite Broadband?

Satellite broadband delivers high-speed internet through satellites rather than depending entirely on terrestrial infrastructure such as optical fibre cables and mobile towers.

Its principal components are:

- **Satellites:** Relay data between users and other parts of the network.
- **User terminals:** Equipment that transmits and receives satellite signals.
- **Ground gateways:** Stations connecting the satellite network with terrestrial internet infrastructure.

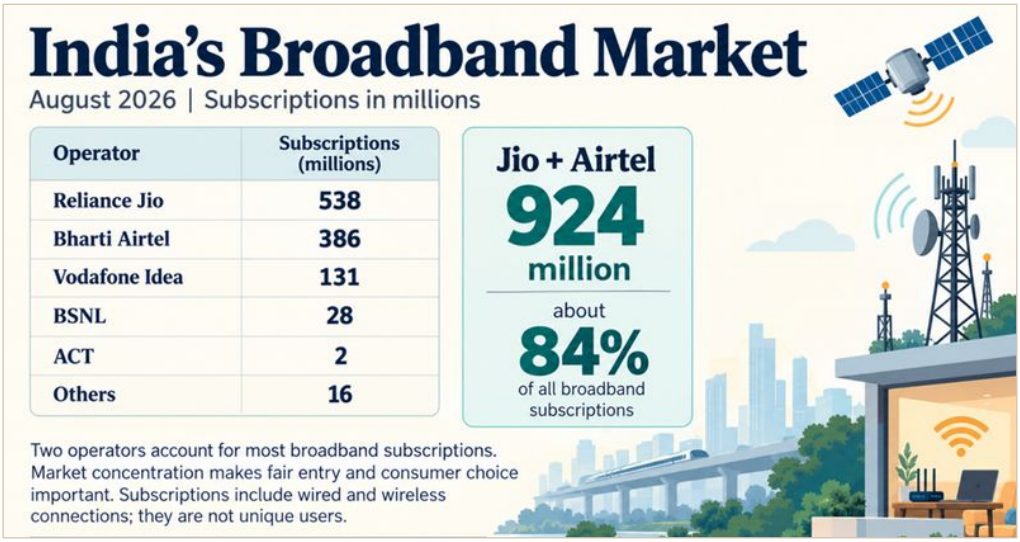
Starlink uses a **Low Earth Orbit (LEO) satellite constellation**—a coordinated network of satellites moving relatively close to Earth.

Feature	LEO satellite networks	Geostationary satellite networks
Position	Closer to Earth; satellites move relative to the ground	Approximately 35,786 km above the equator; appear stationary
Latency	Generally lower because signals travel shorter distances	Generally higher because signals travel farther
Coverage per satellite	Smaller geographical footprint	Larger geographical footprint
Network requirement	Numerous satellites and frequent handovers	Fewer satellites can cover large regions

Latency is the time taken for data to travel through a network. Lower latency improves interactive applications such as video calls.

India's Broadband Market and Competition

India offers a large potential market for satellite internet. However, new providers must compete with established mobile and fibre broadband operators.



Jio and Airtel together account for approximately **924 million subscriptions—about 84% of the reported total**. This indicates significant **market concentration**, where a small number of firms hold a large market share.

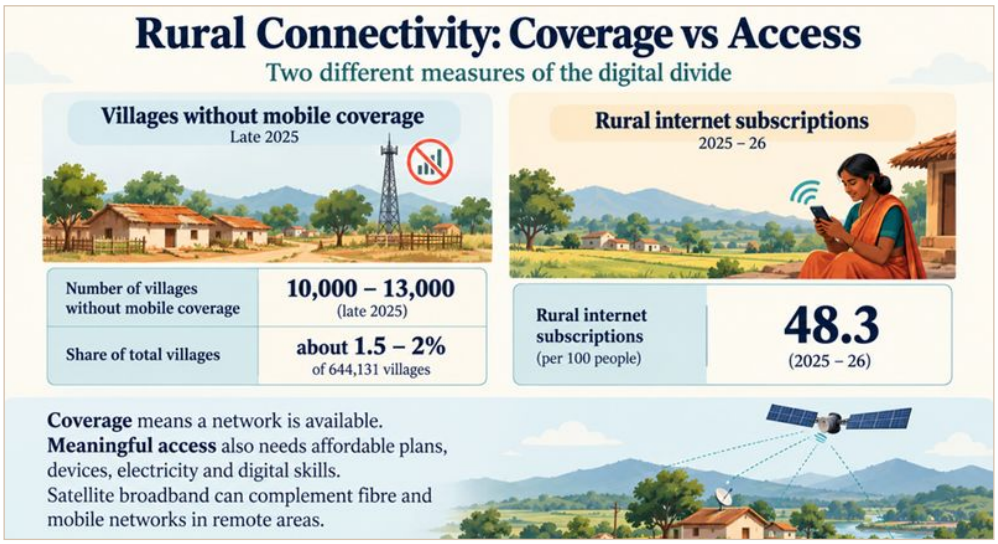
Large operators can achieve economies of scale and undertake extensive infrastructure investment. Nevertheless, transparent entry conditions and effective competition are necessary to protect consumer choice.

High market share alone does not establish anti-competitive conduct. Allegations about barriers to entry require examination based on evidence.

Data distinction: Broadband includes wired and wireless connections. Subscription numbers do not represent unique individuals because one person may have multiple connections.

Bridging the Rural Digital Divide

10,000–13,000 villages lacked mobile coverage in late 2025, approximately **1.5–2% of the stated total of 644,131 villages**. It also reports **48.3 rural internet subscriptions per 100 people in 2025–26**.



These indicators measure different aspects of connectivity:

Concept	Meaning
Network coverage	Availability of a network signal
Subscription density	Number of subscriptions relative to population
Meaningful access	Ability to use affordable, reliable internet with suitable devices and skills

Satellite broadband can serve **mountainous regions, islands, remote settlements and disaster-affected areas**, supporting:

- Online education and telemedicine.
- Access to government services.
- Communication during disruption of terrestrial networks.
- Connectivity for businesses and public institutions.

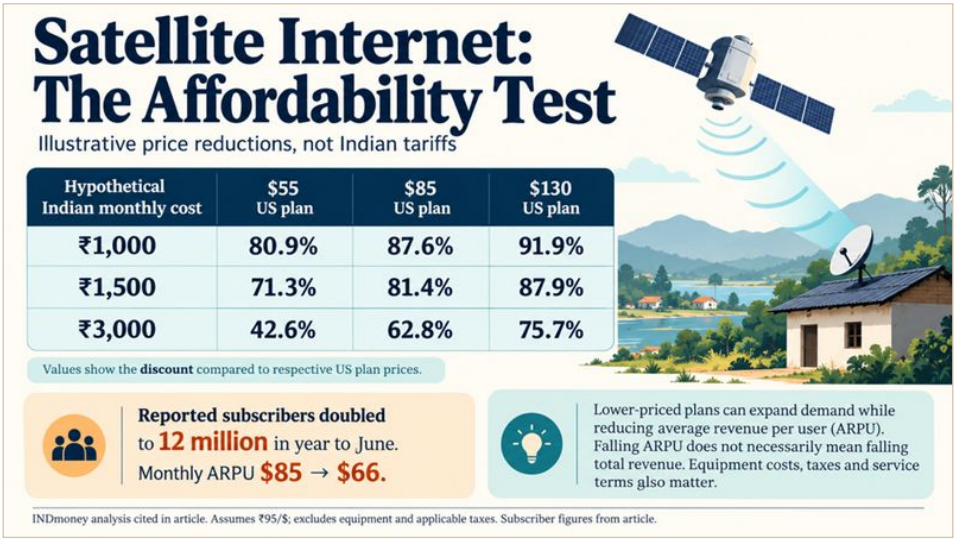
However, **coverage does not automatically ensure adoption**. Equipment costs, subscription charges, electricity availability and digital literacy remain important.

Affordability and Commercial Viability

India is a price-sensitive market. Satellite providers must balance affordable tariffs with the substantial costs of building and maintaining their networks.

According to the article, Starlink's subscribers increased to **12 million in the year to June**, while monthly **average revenue per user (ARPU)** declined from **$85 to $66**.

ARPU measures average revenue earned per subscriber over a specified period. A decline in ARPU can accompany rising total revenue if subscriber growth is sufficiently strong.



Equal advertised speeds do not imply equivalent services. Data limits, latency, reliability, equipment costs and geographical availability also influence value.

Mains question: "Expanding network coverage alone cannot bridge India's digital divide." Discuss the role and limitations of satellite broadband. **(250 words)**

To be continued in Part 2: "Satellite Broadband: Markets, Security and Space Sustainability."